

4. (a) Theories about the Universe have changed over time.

Tick (✓) the boxes next to the names of **two** of these theories.

[2]

Liquid State

☐

Steady State

☐

Loud Bang

☐

Big Bang

☐

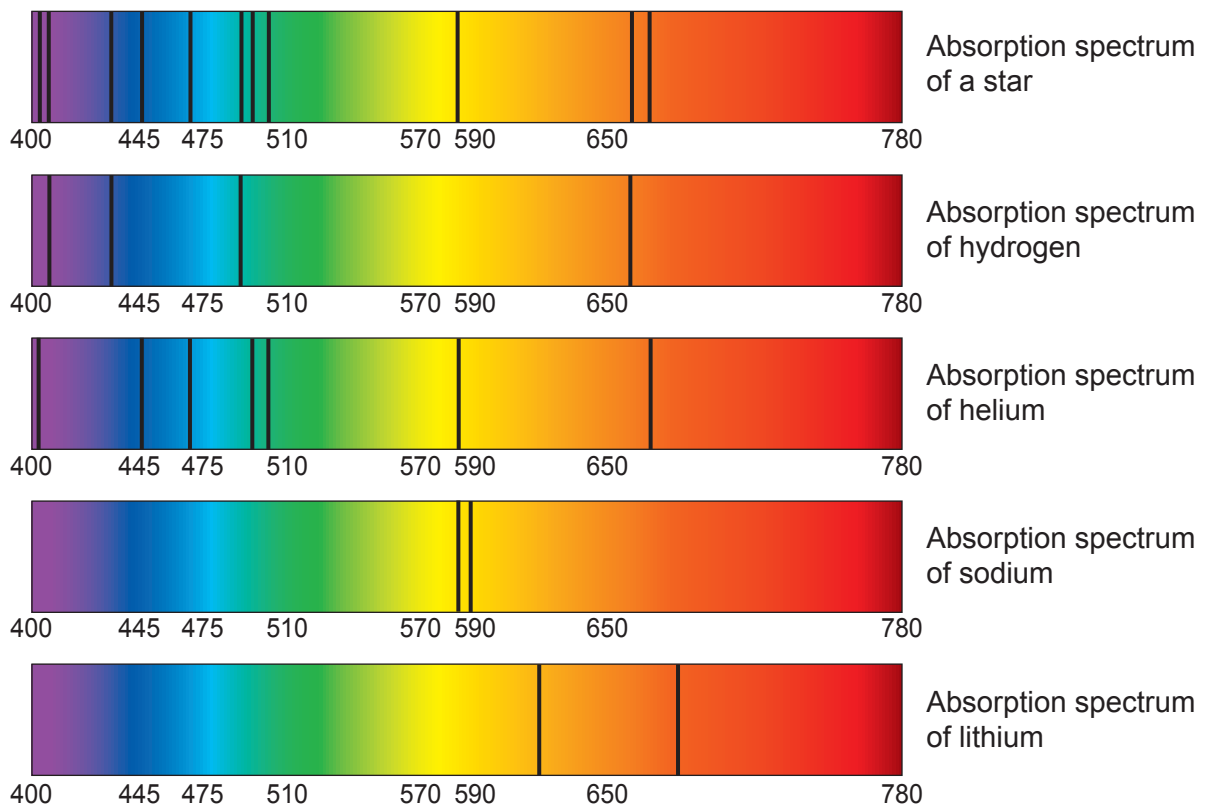
- (b) Images of the Universe are taken using electromagnetic waves.

Name **two** types of electromagnetic wave.

[2]

..... and

- (c) The diagram shows a spectrum from a star and four elements.



Examiner
only

Complete the table by placing a tick (✓) if the element is present in the star. Hydrogen has been done for you.

[1]

Element	Element is present (✓)
hydrogen	✓
helium	
sodium	
lithium	

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(d) The table gives information about the planets in our solar system.

	Mercury	Venus	Earth	Mars	Jupiter	Saturn	Uranus	Neptune
density (kg/m^3)	5427	5243	5514	3933	1326	687	1271	1638
mean distance from the Sun (million km)	58	108	150	228	779	1434	2873	4495
mean speed of orbit (km/s)	47.4	35.0	29.8	24.1	13.1	9.7	6.8	5.4
mean temperature ($^{\circ}\text{C}$)	167	464	15	-65	-110	-140	-195	-200
number of moons	0	0	1	2	79	82	27	14

Use the information in the table to answer the questions below.

(i) Name the coldest **rocky** planet. [1]

(ii) State how many **rocky** planets have moons. [1]

(iii) Name the gas giant that orbits the Sun with the greatest **speed**. [1]

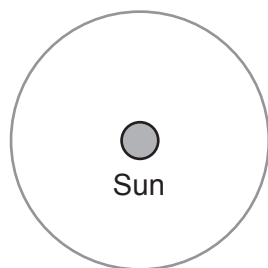
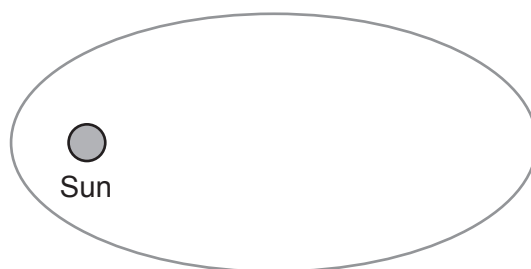
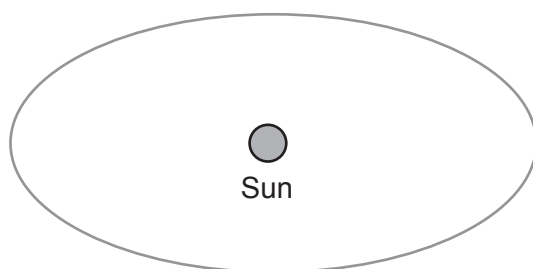
(iv) Eris is a dwarf planet. Its orbit is 10 180 million km away from the Sun.
Shaun says it is found in the asteroid belt between the orbits of Mars and Jupiter.
Explain whether you agree with Shaun. [1]

.....
.....
.....



(e) Tick (✓) the box next to the shape of the orbit of a comet around the Sun.

[1]

☐☐☐

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